



Review article

Genes involved in virulence of the entomopathogenic fungus *Beauveria bassiana*

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ABSTRACT

Pest insects cause severe damage to global crop production and pose a threat to human health by transmitting diseases. Traditionally, chemical pesticides (insecticides) have been used to control such pests and have proven to be effective only for a limited amount of time because of the rapid spread of genetic insecticide resistance. The basis of this resistance is mostly caused by (co)dominant mutations in single genes, which explains why insecticide use alone is an unsustainable solution. Therefore, robust solutions for insect pest control need to be sought in alternative methods such as biological control agents for which single-gene resistance is less likely to evolve. The entomopathogenic fungus *Beauveria bassiana* has shown potential as a biological control agent of insects, and insight into the mechanisms of virulence is essential to show the robustness of its use. With the recent availability of the whole genome sequence of *B. bassiana*, progress in understanding the genetics that constitute virulence toward insects can be made more quickly. In this review we divide the infection process into distinct steps and provide an overview of what is currently known about genes and mechanisms influencing virulence in *B. bassiana*. We also discuss the need for novel strategies and experimental methods to better understand the infection mechanisms deployed by entomopathogenic fungi. Such knowledge can help improve biocontrol agents, not only by selecting the most virulent genotypes, but also by selecting the genotypes that use combinations of virulence mechanisms for which resistance in the insect host is least likely to develop.

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1. Introduction

Arthropod pests are a nuisance for human health and global production of crops used for food, biofuel, and fibers. Different insect species act as vectors that transmit lethal diseases, including malaria and dengue. Malaria has the highest death rate of all vector-borne diseases, with 584,000 estimated mortalities in 2013 (WHO, 2014). On the other hand, dengue is the fastest increasing vector-borne disease, with a global estimate of 390 million new infections annually (Bhatt et al., 2013). Control of disease vectors is an important means to prevent infection and spreading of diseases because these vectors can be the most vulnerable or preventable stage of human disease development (Zaim and Guillet, 2002). Other insect pests cause serious damage to crops grown throughout the world. Reduction of the annual global crop yield by insect pests was estimated to amount to 18% a decade ago, apart from loss of stored grains (Bergvinson and García-Lara, 2004; Oerke and Dehne, 2004). A more recent study estimated a loss of about 8% of the major crops in Brazil due to pests, in spite of control measures (Oliveira et al., 2014).

Though chemical pesticides are widely applied in agriculture since many decades, several complications undermine their applicability, such as acquired insecticide resistance, serious cases of pesticide poisoning, and environmental pollution (Oliveira et al., 2014; Zaim and Guillet, 2002). There is a growing urgency to develop alternative pesticides, and biopesticides are of promising potential. Various entomopathogenic fungi, such as *Beauveria bassiana* and *Metarhizium spp.*, are currently used as biological control agents for different arthropods, both to fight the dispersal of diseases and to reduce the loss of crops or harvested produce (de Faria and Wraight, 2007).

The cosmopolitan ascomycete fungus *B. bassiana* is capable of exploiting a range of environments, including soil, plants and insects; the fungus is able to live as a saprophyte in the soil, as

an endophyte in plants, or as an entomopathogen affecting a broad range of arthropods (Boomsma et al., 2014). It can reproduce sexually in a teleomorphic state called *Cordyceps bassiana* (Sung et al., 2006). Conidia can start an infection upon contact with a compatible insect host, then colonize the host and ultimately kill it. *B. bassiana* insect hosts include crop pests (e.g. whiteflies and borers, (Brownbridge et al., 2001; Cruz et al., 2006)), ecologically hazardous pests (e.g. termites, (Jones et al., 1996)) and insect disease vectors, including mosquitoes and ticks (Blanford et al., 2005; Kirkland et al., 2004). Its wide host range makes *B. bassiana* a suitable organism to control a variety of pests (Zhang et al., 2011b), as long as specific delivery tools are used to target only the pest species.

To further assess whether a biocontrol approach that makes use of entomopathogenic fungi is a robust and sustainable solution to arthropod pests, we need a better understanding of the life history of the biocontrol agent as well as the mechanisms and modes of action in killing the pest. With the recent availability of the whole genome sequence of *B. bassiana* (Xiao et al., 2012), rapid progress can be made in understanding the genetics that constitute virulence toward insects. This review provides an overview of our current knowledge of the genes contributing to virulence in entomopathogenic fungi, with an emphasis on *B. bassiana*. Current insights in virulence mechanisms are discussed, implicating possibilities to increase the efficiency and killing speed of *B. bassiana* strains in order to improve their potential as a biopesticide in the fight against crop loss and disease dispersal by insects.

2. Insect infection cycle by *B. bassiana*

This review focuses on the infection cycle of *B. bassiana* in insects, which can be divided into seven steps (Fig. 1). To discuss the life cycle of *B. bassiana* is beyond the scope of this article, but see Boomsma et al. (2014) for a complete overview. To enable

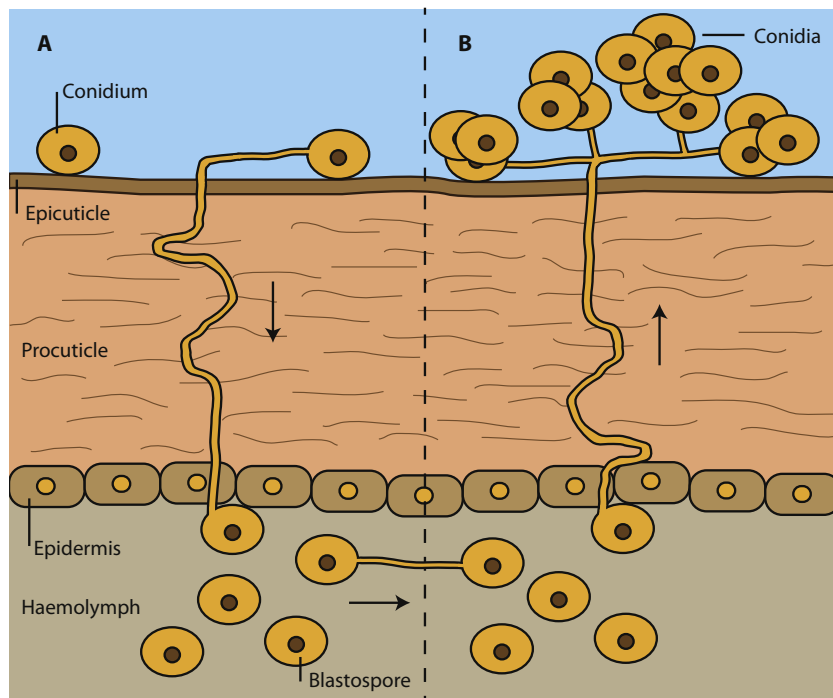


Fig. 1. Infection cycle of *Beauveria bassiana* in insects (adapted from Clarkson and Charnley, 1996). (A) A conidium must adhere to the insect host to induce germination (step 1). This is followed by germination and production of a germ tube (step 2). Mechanical pressure and secretion of enzymes are employed to breach the cuticle (step 3). The fungus switches to blastospore growth to colonize the host hemocoel (step 4), where it secretes toxins and feeds on sugars in the hemolymph (step 5). The blastospores evade or overcome host immune responses and secrete toxins that facilitate killing of the host (step 6). (B) After the host has died, the fungus breaches the cuticle again from the inside and sporulates on the cadaver (step 7). The arrows indicate the direction of fungal growth.

Table 1

Overview of discussed *B. bassiana* genes involved in virulence, in order of the main affected infection step (1 = host adhesion, 2 = germination, 3 = cuticle degradation, 4 = growth as blastospores, 5 = host colonization and killing, 6 = immune response interactions, 7 = hyphal extrusion and conidiation).

Gene name	Step	Characteristics	Knock-out mutant phenotype	References
hyd1	1	Hydrophobin	Reduced virulence	Zhang et al. (2011b)
hyd2	1	Hydrophobin	Reduced host adherence	Zhang et al. (2011b)
Bbhog1	1, 2, 6, 7	MAP kinase	Reduced virulence, conidial yield, host adherence, appressoria and stress-tolerance	Zhang et al. (2009)
Bbmpk1	1, 2, 7	MAP kinase	No appressoria, reduced cuticle breaching, host adherence and conidiation	Zhang et al. (2010)
Bbsnf1	2, 4, 7	Protein kinase	Reduced virulence, acidification, conidiation and blastospore yield	Wang et al. (2014c)
Bbcyp52x1	2, 3, 7	Cytochrome P450	Reduced virulence, germination and cuticle breaching	Zhang et al. (2012)
cdep1	3, 7	Subtilisin-like protease	–	Fang et al. (2008)
Pr1, Pr2	3, 7	Subtilisin-like protease	–	Joshi et al. (1995), Dias et al. (2008)
Bbchit1	3, 7	Chitinase	Overexpression increased infection efficiency	Fang et al. (2005)
Chi1, Chi2, ChsA2	3, 7	Chitinase	–	Liu et al. (2013)
cdc25	4, 6, 7	Regulates Cdk1 activity	Altered cell morphology, reduced virulence, conidiation and stress-tolerance	Qiu et al. (2015)
wee1	4, 6, 7	Regulates Cdk1 activity	Altered cell morphology, reduced virulence, conidiation and stress-tolerance	Qiu et al. (2015)
Fkh2	4, 6, 7	Transcription factor	Altered cell morphology, reduced virulence, conidiation and stress-tolerance	Wang et al. (2015)
Bbagt1	5, 7	α -glucose transporter	Reduced virulence, germination and conidial yield	Wang et al. (2013c)
Bbmpd	5, 6	Mannitol-1-phosphate dehydrogenase	Reduced virulence, stress-tolerance and trehalase expression	Wang et al. (2012)
Bbmt	5, 6	Mannitol dehydrogenase	Reduced virulence, stress-tolerance and trehalase expression	Wang et al. (2012)
BbCreA	3, 5, 7	Regulates CreA activity	Reduced virulence, conidiation and nutrient uptake	Luo et al. (2014)
Bbcsa1	5	Calcium sensor, acidification regulator	Reduced extracellular acidification	Fan et al. (2012b)
Bbbeas	5	Beauvericin toxin	Reduced virulence and beauvericin production	Xu et al. (2008)
Bbbsls	5	Bassianolide toxin	Reduced virulence and bassianolide production	Xu et al. (2009)
(geneID not known)	5	Bassiactridin	(Purified protein showed activity specific to locusts)	Quesada-Moraga and Vey (2004)
ktr1	6	α -1,2-mannosyltransferase	Reduced conidial thermotolerance	Wang et al. (2014b)
ktr2, ktr4	6, 7	α -1,2-mannosyltransferase	Reduced conidiation and stress-tolerance	Wang et al. (2014b)
ras1, ras2, ras3	2, 6, 7	GTPase, signal transduction	Reduced virulence, germination, conidiation, stress-tolerance	Xie et al. (2013), Guan et al. (2015)
Bbac	6, 7	cAMP signaling	Reduced virulence, conidiation and stress-tolerance	Wang et al. (2014a)
cdc14	4, 6, 7	Cytokinesis	Reduced blastospore production, conidial yield and stress-tolerance	Wang et al. (2013a)
Bbmsn2	3, 6, 7	Transcription factor	Reduced conidial yield and stress-tolerance	Liu et al. (2013)
sod1-5	6	Superoxide dismutases	Reduced stress-tolerance	Li et al. (2014)
vcx1A-E	6	Vacuolar calcium exchangers	Reduced virulence and stress-tolerance	Hu et al. (2014)
catA-D, catP	6	Catalase	Reduced virulence, stress-tolerance	Wang et al. (2013d)
trx1-6	6	Thioredoxin antioxidants	Reduced virulence, germination, conidiation and stress-tolerance	Zhang et al. (2015)
Mdr1, Mrp1, Pdr1, Pdr2, Pdr5	6	ABC transporters	Reduced stress-tolerance	Song et al. (2013)
CnA1, CnA2, CnB, Crz1	6, 7	Calcineurin-mediated phosphatase	Reduced virulence, stress-tolerance and conidiation	Li et al. (2015)
Bbrgs1	7	G-protein signaling	Reduced conidial yield, thermotolerance and viability	Fang et al. (2008)
BbGPCR3	4, 7	G-protein coupled receptor	Reduced conidiation, virulence and transition to blastospores	Ying et al. (2013)
Bbslt2	7	MAP kinase	Reduced conidial yield and viability, cell growth and cell wall sensitivity	Luo et al. (2012)
BbmrA	7	Putative methyltransferase	Reduced germination, growth, conidiation and virulence	Qin et al. (2014)
Bbgas1	2, 7	Cell-wall anchored enzyme	Reduced virulence and thermotolerance	Zhang et al. (2011a)
Bbjen1	7	Carboxylate transporter	Reduced virulence, increased conidial yield	Jin et al. (2010)
Bbpml1	7	ATPase, signaling	Reduced conidiation	Wang et al. (2013b)

infection, conidia must adhere to the host cuticle (step 1) and subsequently germinate to form a germ tube and in particular cases an appressorium (2). The cuticle can then be breached using a combination of mechanical pressure and enzymatic degradation (3), enabling hyphae to grow through the cuticle and invade the insect hemocoel, where they switch to growth as single-celled blastospore (4). These cells circulate freely through the hemocoel, where they exploit the hemolymph for nutrition and secrete toxins that eventually kill the host (5). Successful invasion requires that the fungal cells evade or overcome immune responses (6), both during growth through the cuticle and especially in the hemocoel. When the host is killed, the cuticle must be breached again from the inside to allow escape from the insect body and sporulate on the cadaver (7), enabling conidia to be dispersed and start a new infection. We will discuss each of the steps in more detail and

describe what is currently known about genes that play a role in each of the steps of the *B. bassiana* infection cycle (summarized in Table 1). Most of this information results from the phenotypic characterization of targeted knock-out mutants. Due to the complex life history of *B. bassiana*, many of the virulence genes studied are involved in more than one step of the infection cycle, or they are relevant for fundamental processes such as conidiation and cell wall formation. In cases where the deletion of a candidate virulence gene visibly affects the growth rate and cellular development of the fungus *in vitro*, it is difficult to draw conclusions about the specific function of a gene in the infection process, since developmental defects may impinge on the virulence in an indirect manner. For completeness, we have nevertheless included information on pleiotropic developmental genes in our list of “virulence genes”.

2.1. Host adherence

The primary route for successful infection is adherence of a conidium to the host. *B. bassiana* aerial conidia have a hydrophobic coating and adhere to an insect cuticle through hydrophobic surface interactions. Passive adhesion can be mediated by hydrophobins (Holder and Keyhani, 2005). *B. bassiana* expresses two hydrophobin genes, *hyd1* and *hyd2*, encoding proteins that accumulate and covalently aggregate on the surface of conidia. The $\Delta hyd1 \Delta hyd2$ double knock-out mutants showed reduced conidial adhesion and decreased virulence, though both genes have different functions. Whereas the $\Delta hyd2$ knock-out mutant conidia showed reduced surface adhesion but unchanged virulence, the $\Delta hyd1$ knock-out mutant displayed decreased virulence but unchanged surface adhesion (Zhang et al., 2011b).

Apart from hydrophobicity, electrical charge of the conidia is thought to affect adhesion. Conidial surfaces have a net negative charge that attracts them toward positively charged surfaces. However, neutralization of the conidial surface charge did not inhibit binding of conidia to the insect cuticle (Boucias et al., 1988).

The compositions of cuticle and conidial surfaces influence adhesion potential. For example, fatty amides in the cuticle of the insect host *Liposcelis bostrychophila* reduce conidial adhesion, by decreasing electrical charge and hydrophobicity (Lord and Howard, 2004). In a similar way, the composition of proteins and sugars on the conidial surface can affect their adhesion through charge and hydrophobicity. MAD1 and MAD2 are two adhesin-like proteins identified in *Metarhizium anisopliae*. These proteins, present on the conidial surface, were found to be required for adhesion to insect cuticle and plant epidermis, respectively. In addition, MAD1 influences germination (step 2), blastospore formation (step 5) and, ultimately, virulence (Wang and St. Leger, 2007a). Homologs have been found in *B. bassiana*, though their functions remain to be elucidated (Xiao et al., 2012).

2.2. Germination and penetration peg development

Once aerial conidia have adhered to the host cuticle, they can germinate when the right conditions are met. These conditions include favorable temperature and humidity, and the presence of an exogenous carbon source (Bidochka et al., 2000). Germinated conidia grow germ tubes, that develop into a penetration peg that perforates the cuticle to invade the host. Little is known about genes that are involved in host sensing and regulating the differentiation of penetration pegs. In some strains, it was reported that germings differentiate into appressoria, although the formation of an appressorium structure is not required for a successful infection (Wagner and Lewis, 2000). In *B. bassiana* strain Bb0062, the MAP kinase gene *Bbmkp1* was shown to be required for appressorium formation and penetration on cicada wings (Zhang et al., 2010). Also, the disruption of the MAP kinase gene *Bbhog1* resulted in reduced frequency of appressorium formation and reduced virulence (Zhang et al., 2009). MAP kinases affect a number of processes in entomopathogenic fungi, including conidial adhesion (step 1), appressorium formation and virulence. Regulators, such as MAP kinases, affect the expression of several downstream genes, and their disruption has pleiotropic effects. Such studies therefore do not elucidate which target genes are specifically involved in appressorium formation. More targeted studies are required to unravel how appressoria (if they are formed) contribute to virulence.

Although appressorium formation in *B. bassiana* is not prevalent and their role in virulence is not clear, the formation of an appressorium in *Metarhizium* species is clearly established and several studies have been performed on genes involved in their formation.

The gene *mpl1* (*Metarhizium* perilipin-like protein) was shown to function in the storage of lipids in droplets, especially in appressoria. Delayed appressorium formation and reduced turgor pressure were observed in knock-out mutants in *Mapl1*, a gene highly expressed in appressoria (Luo et al., 2013). Appressoria of the $\Delta mpl1$ knock-out mutant showed strongly reduced turgor pressure and impaired ability to breach the cuticle (Wang and St. Leger, 2007b). Knock-out mutants in the response regulator *Mrskn7* also were impaired in virulence, resulting from inability to form appressoria (Shang et al., 2015). Deletion of the gene *Magas1* resulted in reduced cuticle penetration, though appressorium formation was unaffected (Cao et al., 2012). A recurring finding in these studies was that the reduction of virulence was only observed upon topical application, but not upon intrahemocoel application of conidia, indicating fundamental roles of these genes in cuticle penetration.

2.3. Enzymatic cuticle degradation

Germ tubes on the cuticle are subject to conditions of environmental stress, such as high temperature and UV exposure, and to insect defense responses in the cuticle. Accelerated breaching of the cuticle may reduce the time of exposure to these conditions and thereby increase the chance of successful infection, thus positively contributing to the virulence of the fungus (Zhang et al., 2010). Penetration of the cuticle is not enabled solely by mechanical pressure, but is also facilitated by simultaneous enzymatic degradation of the cuticle.

The cuticle consists mainly of a framework of chitin, supporting proteins and lipids. The proportions of these three components in the migratory grasshopper *Melanoplus sanguinipes* were circa 30%, 61% and 7.4%, respectively (Bidochka and Khachatourians, 1992). Entomopathogenic fungi secrete various enzymes to degrade these cuticle components, mainly proteases and chitinases (Charnley and St. Leger, 1991). The cuticle is degraded both to obtain nutrients and to weaken the structural barrier, enabling host invasion.

The outermost, waxy layer of the epicuticle is rich in lipids, including fatty acids and hydrocarbons. The hydrocarbons present in the epicuticle mainly include alkenes, methyl-branched chains and *n*-alkanes, the latter of which are sufficient to serve as a carbon source for the fungus. Insects can change the hydrocarbon content of the cuticle in response to fungal infection, which may influence the success of infection, as different hydrocarbons can either inhibit or stimulate conidial adhesion, germination and growth (Pedrini et al., 2013).

A cytochrome P450 gene, *Bbcyp52x1*, was shown to function in enzymatic breakdown of lipids in the waxy layer. Conidia of a $\Delta Bbcyp52x1$ knock-out mutant showed impaired germination on grasshopper wings. This knock-out mutant also showed reduced virulence on the Greater Wax moth *Galleria mellonella* upon topical application of conidia, though not upon intrahemocoel injection. Similar to *Magas1* (step 2) this suggests that the *Bbcyp52x1* gene promotes virulence specifically via its function in cuticle penetration (Zhang et al., 2012). *B. bassiana* is known to secrete lipases (Leopold and Samšínáková, 1970), which are thought to play a role in degradation of the waxy layer. However, a study by Feng (1998) indicated that the total lipase activity of various strains had little correlation to their virulence.

As indicated above, proteins are the main component of the insect cuticle and assays with cuticles from which proteins had been removed indicated that proteins are the most important nutrient source for germination and hyphal growth on the cuticle (Bidochka and Khachatourians, 1992). Therefore, not surprisingly, extracellular proteases are important for cuticle breakdown by

entomopathogenic fungi. Specifically in *B. bassiana*, subtilisin-like (Pr1) and trypsin-like (Pr2) cuticle-degrading proteases were shown to be involved, as these genes were induced by the presence of cuticle compounds of different insects (Joshi et al., 1995). The Pr1 gene *cdep1* also showed involvement in cuticle breakdown in *B. bassiana* (Fang et al., 2008).

The insect chitin framework can be degraded by secreted chitinases. Several *B. bassiana* chitinases were studied with respect to their function in this process. Fang et al. (2005) experimentally supported the involvement of a *B. bassiana* chitinase in virulence on aphids, *Myzus persicae*. Overexpression of chitin gene *Bbchit1* increased virulence, both by increasing the infection efficiency and accelerating the infection (Fang et al., 2005). Recently, Liu et al. (2013) reported that a $\Delta Bbmsn2$ knock-out mutant displayed lower transcript levels of the three chitinase genes *Chi1*, *Chi2* and *ChsA2*, and showed reduced virulence (Liu et al., 2013).

A study into the protein kinase gene *Bbsnf1* showed a reduction of virulence in the $\Delta Bbsnf1$ knock-out mutant, and the reduction was more pronounced upon topical application than upon intra-hemocoel injection. This suggests a role for this regulator in cuticle penetration, though the exact mode of action of BbSNF1 is currently not known (Wang et al., 2014c).

2.4. Transition to blastospore growth

Hyphae that have successfully grown through the cuticle and reached the hemocoel can switch to a yeast-like growth mode. The fungus forms single-cell hyphal bodies called blastospores (*in vivo* blastospores), which have a unique brush-like outermost structure (Wanchoo et al., 2009). This is a cryptic growth form, and this unique structure is thought to help blastospores avoid detection by the host immune system. Blastospores proliferate via budding and use the circulating hemolymph as a 'vehicle' for colonization of the host as well as a source of nutrition (Lewis et al., 2009).

The identities of genes specifically regulating the developmental transition from hyphal growth to blastospores remains unknown in *B. bassiana*, though certain regulatory genes of many biological processes were shown to be involved in blastospore development. For example, analysis of the role of the *wee1* gene and the *cdc25* gene (which both regulate the cell cycle-required cyclin-dependent kinase 1 Cdk1), showed that blastospore size was reduced in the knock-out mutants in these genes (Qiu et al., 2015). The forkhead transcription factor *Fkh2* is also involved in the regulation of the cell cycle, and the deletion of *Fkh2* produced smaller blastospores as well, but twice as many blastospores compared the wild-type (Wang et al., 2015). Moreover, disruption of the *Bbsnf1* gene, a protein kinase involved in fungal development, led to lower blastospore yield (Wang et al., 2014c).

2.5. Colonization and killing of the host

Different virulence determinants play a role during host colonization, both for growth in the insect and to mediate the death of the host. During infection, the insect body is the source of all nutrition for the fungus, and especially the disaccharide trehalose, which is the most abundant carbohydrate in the insect hemolymph. The fungus either secretes trehalase to hydrolyze the circulating trehalose and take up the resulting glucose, or imports trehalose to process it intracellularly (Xia et al., 2002). The α -glucoside transporter gene *Bbagt1* was shown to be necessary for trehalose uptake, and its deletion resulted in reduced virulence (Wang et al., 2013c). In addition to trehalose utilization, it is likely that virulence is correlated with the metabolism and biosynthesis of other sugars. One of the few detailed studies on this topic shows that the mannitol dehydrogenase genes *Bbmptd* and

Bbmtd, involved in mannitol biosynthesis, are also required for full virulence. Interestingly, the knock-out mutant strains of each gene also showed a decreased trehalase expression (Wang et al., 2012). Another example is the *BbCreA* regulator that plays an important role in nutrient utilization. The $\Delta BbCreA$ knock-out mutant showed a reduced growth in carbohydrate substrates, improper assimilation of some amino acids as well as reduced virulence and conidiation (Luo et al., 2014).

During infection, the fungal cells encounter different pH values, depending on the environment. The pH inside a host varies between insect species and tissues, and it changes with age (Schultz and Lechowicz, 1986). Secreted enzymes have a pH at which their activity is optimal, and therefore gene expression must be adapted to the ambient pH. In *Metarhizium anisopliae*, several genes encoding cuticle degrading enzymes showed highest transcript levels around their optimal pH (St Leger et al., 1998). The pH-dependent regulation of expression is mediated by response elements in the genome. The transcription factor *BbpacC* is one such element that enables *B. bassiana* to adapt its gene expression to ambient pH (Peñalva et al., 2008). Alkaline pH strongly induces *BbpacC*, leading to upregulation of alkaline-expressed genes and repression of acid-expressed genes (Zhou et al., 2014).

The ambient pH can also be regulated by secretion of compounds that acidify or alkalinize the surrounding microenvironment (St Leger et al., 1999). Progress of the infection can be stimulated by acidification of the host tissue and hemolymph. To this end the fungus secretes acidic metabolites such as oxalic acid (Bidochka and Khachatourians, 1991). A factor involved in acidification is the *Bbcsa1* gene (calcium sensor and acidification regulator). Deletion of the *Bbcsa1* gene resulted in reduced acidification accompanied by reduced virulence (Fan et al., 2012b).

In addition to acidifying compounds, the fungus can secrete toxic metabolites to speed up the death of the host. *B. bassiana* produces beauvericin, which has insecticidal, antifungal and cytotoxic properties. Beauvericin synthase (*Bbbeas*) was found to be important, though not essential for virulence, as different insect hosts infected with the $\Delta Bbbeas$ knock-out mutant showed significantly reduced mortality rates (Xu et al., 2008). The cytotoxic cyclodepsipeptide bassianolide synthase (*Bbbsls*) has been detected in cadavers of silkworms killed by *B. bassiana*, suggesting that production of this toxin coincides with infection and insect death. $\Delta Bbbsls$ knock-out mutant strains showed decreased virulence on different hosts (Xu et al., 2009). Another study showed secretion of the toxin bassiacridin, which increases virulence specifically on locusts (Quesada-Moraga and Vey, 2004).

2.6. Interactions with the insect immune system

From cuticle adhesion onwards, the fungus has to survive the insect immune responses and including defense mechanisms such as toxins, protease inhibitors or other infection-inhibiting compounds that can be present in the cuticle (for a review on this topic, see Ortiz-Urquiza and Keyhani (2013)). The fungus must either evade or overcome the host defense mechanisms (e.g. by secreting compounds that modulate the insect response or by morphological structures in blastospores). Details are scarce on the genetic regulation of the responses of entomopathogenic fungi to the host defense mechanisms.

The insect innate immune responses are regulated by the Toll and Imd (Immune deficiency) pathways. The Toll signaling pathway is activated by the ligand Spätzle, which can be activated via serine protease cascades by microbial compounds such as fungal β -1,3-glucans (Valanne et al., 2011). The insect immune system can be divided in a humoral and a cellular immune response. A humoral response includes the release of a range of anti-microbial

peptides (AMPs) into the hemolymph (Tanji et al., 2007), the release of reactive oxygen species (ROS) and a complex enzymatic cascade that regulate clotting or melanization (Jiravanichpaisal et al., 2006). Activation of a melanization response leads to the deposition of melanin on the surface of parasites or wound surface. This results in the release of ROS and encapsulation of the fungal cells (Pham et al., 2007; Yassine et al., 2012). Consequently, mortality of *Anopheles gambiae* larvae infected with the mold *Lagenidium giganteum* was reduced by melanization of germ tubes (Golkar et al., 1993).

In contrast, the cellular response is governed by different types of hemocytes, which can encapsulate, phagocytize and kill invading cells (Pham et al., 2007). Blastospores of *B. bassiana* phagocytized by *Spodoptera exigua* granulocytes were found to escape by growing a germ tube out of the encapsulation, where they started budding to form new blastospores (Hung et al., 1993). In addition, hemocyte production in infected larvae was reduced (caused by immunosuppressive compounds) and the hemocytes that were produced showed reduced efficacy in encapsulating fungal cells (Hung and Boucias, 1992). It is unknown which fungal genes contribute to overcoming immune responses in the insect host, but it is tempting to speculate the involvement of effector proteins (small secreted cysteine-rich proteins) that are able to suppress the immune response, as extensively studied in plant pathogenic fungi (Stergiopoulos and de Wit, 2009). Next to the escape reaction, hyphal bodies seem able to simply evade recognition by the absence of antigenic compounds on the cell surface, such as chitin and galactomannan (Pendland et al., 1993). In *M. anisopliae*, the MCL1 protein was found to mask compounds on the cell surface and thereby enables hyphal bodies to avoid detection by hemocytes in the insect hemolymph. The hydrophilic properties and negative charge of MCL1 proteins reduce the adhesion of hyphal bodies to hemocytes (Wang and St Leger, 2006). No homolog of MCL1 has been detected in the *B. bassiana* genome and it is unknown whether a different protein fulfills an analogous function to the MCL1 protein.

Insect hosts can increase hemolymph ROS concentrations in response to infection. This increases survival after infection, as shown in *A. gambiae* infected with bacteria or *Plasmodium* (Molina-Cruz et al., 2008). The host employs enzymes to protect itself against ROS, among which are catalases and thioredoxin peroxidases to detoxify H₂O₂, and superoxide dismutases (SODs) to detoxify O₂⁻ superoxide (Molina-Cruz et al., 2008). For successful infection, the invading fungus must likewise protect itself against exposure to ROS and concomitant cellular stress. A number of genes were found to be involved in ROS protection and stress tolerance, which are not only relevant for infection but also for other life stages of the fungus. Those genes are mostly regulators that determine virulence by affecting multiple interconnected downstream processes, among which are conidiation, multi-stress responses, germination, and growth.

As examples of such regulators, the genes *Ras1*, *Ras2* and *Ras3* influence virulence by affecting multi-stress tolerance, germination and growth, among others (Guan et al., 2015; Xie et al., 2013). The adenylate cyclase gene *Bbac*, operating in cAMP signaling, was also found to contribute to virulence, by regulating multi-stress responses and conidiation. The $\Delta Bbac$ knock-out mutant was highly sensitive to stress by high osmolarity, oxidation, cell wall damage and different chemicals. *Bbac*, similar to other regulators, seems to improve tolerance to oxidation by positively regulating antioxidation (Wang et al., 2014a). All these genes have regulatory functions and more detailed studies are required to reveal their downstream target genes and specific roles.

Additional interesting examples of virulence genes functioning in stress responses are calcineurin subunits *CnA1*, *CnA2* and *CnB*

and their downstream transcription factor *Crz1* (Li et al., 2015), the required for cytokinesis gene *Cdc14* (Wang et al., 2013a), the zinc finger transcription factor *Bbmsn2* (Liu et al., 2013), superoxide dismutases *sod1-5* (Li et al., 2014), vacuolar calcium exchangers *vcx1A-E* (Hu et al., 2014), catalases *catA-D* and *catP* (Wang et al., 2013d), the α -1,2-mannosyltransferases *Ktr1*, *Ktr2* and *Ktr4* (Wang et al., 2014b), the thioredoxin antioxidants *trx1-6* (Zhang et al., 2015), and the ATP-binding cassette (ABC) transporters *Mdr1*, *Mrp1*, *Pdr1*, *Pdr2*, and *Pdr5* (Song et al., 2013). For a more specific review on factors contributing to (environmental) stress tolerance in entomopathogenic fungi, see Ortiz-Urquiza and Keyhani (2014).

2.7. Hyphal extrusion and conidiation

When the nutrients in the host are depleted and the host has died, the fungus must grow out of the insect to produce and disperse its conidia (Fig. 1B). Blastospores still circulating in the hemocoel must revert to growth as hyphae, which have to breach the cuticle again – this time from the inside out – to enable hyphal extrusion followed by conidiation on the host surface. It is likely that the same factors are involved here as in breaching the cuticle from the outside, as discussed in infection step 2.

In the previously discussed $\Delta Bbmkp1$ knock-out mutant, the expression of several protease and chitinase genes was unaltered but the fungus failed to breach the cuticle from the inside and did not proceed to conidiation. This indicates that BbMPK1 has a vital role not only for appressorium formation and conidiation but also is crucial for penetration of the insect cuticle both from the outside inwards and from the inside outwards (Zhang et al., 2010). Similar indications that the same factors are involved in inward and outward cuticle penetration were made in *Metarhizium acridum* for the *MaPls1* gene (Luo et al., 2013). Hyphae that have escaped the insect body can proceed to the production of conidia. This process is induced by sensing of favorable environmental conditions via G protein-mediated signaling in filamentous fungi. Regulators of G-protein signaling (RGS) are also involved in other processes, such as germination, stress tolerance and secretion of toxins (Fang et al., 2008). The RGS gene *Bbrgs1* was shown to be involved in conidiation, as gene disruption reduced conidial yield, as well as conidial viability and thermotolerance. Remarkably, virulence was unaffected both upon topical application and intrahemocoel injection (Fang et al., 2008). Involvement of RGS genes in conidiation of *Metarhizium* has also been confirmed (Fang et al., 2007). In addition, a G-protein coupled receptor (GPCR) *BbGPCR3* was shown to be involved in conidiation, nutrient sensing and stress response. A $\Delta BbGPCR3$ knock-out mutant showed reduced conidiation and transition to blastospores, and a decreased virulence in topical applications (Ying et al., 2013).

Genes specifically acting in conidiation are not well studied in *B. bassiana*, but the involvement of various regulators has been reported. Effects on conidiation often manifested in a reduction of conidial yield and viability, and/or altered morphology of conidia-bearing aerial hyphae. Studies on regulatory genes indicate that conidiation is connected with other developmental processes. Reduced conidiation, along with disturbance of other developmental processes, was observed upon disruption of the MAP kinase *Bbslt2* (Luo et al., 2012) and the previously discussed *Bbmkp1* (Zhang et al., 2010) and *Bbhog1* (Zhang et al., 2009). The cell wall anchored enzyme *Bbgas1* proved to be involved in conidial thermotolerance and virulence. Conidia of the $\Delta Bbgas1$ knock-out mutant showed reduced virulence to *G. mellonella* upon both topical application and intrahemocoel injection, and heat-shock treated conidia showed decreased ability to germinate (Zhang et al., 2011a). The putative methyltransferase *BbmtrA* was shown to be involved in conidial viability, fungal growth and virulence

(Qin et al., 2014). The carboxylate transporter *Bbjen1* also is involved in conidiation and virulence, as the $\Delta Bbjen1$ knock-out mutant displayed reduced virulence but increased conidial yield (Jin et al., 2010).

More regulators are known to affect conidiation. Some examples are the previously discussed *Bbac* (Wang et al., 2014a) and *Bbmsn2* (Liu et al., 2013), a knock-out mutant of the gene *A43* had a reduced conidial yield (Wu et al., 2008) and the P-type Ca^{2+} -ATPase *Bbpmr1* regulates conidiation by functioning in different signaling cascades (Wang et al., 2013b).

Studies on regulators make it clear that different developmental processes are connected via signaling pathways. One study established the role of the *Ohmm1* gene, which was downregulated in $\Delta Bbhog1$. A $\Delta Ohmm1$ knock-out mutant showed increased virulence and oxidative stress tolerance, and reduced conidiation (He et al., 2015). Additional studies that focus on the downstream targets of regulators are required to reveal the roles of specific genes and to obtain a more detailed understanding of conidiation and its impact on virulence in *B. bassiana*.

3. Conclusions and perspective

Infection of an insect by *B. bassiana* comprises numerous complex developmental transitions, which require intricate genetic regulation. Specific roles of a number of regulatory genes in *B. bassiana* have been established, while the roles of others have yet to be unraveled. A better understanding of the network of genes that contribute to virulence could ideally be used for selecting strains with the most virulent genotype. Insight in virulence genes (see Table 1 for an overview) can contribute to improving the efficiency of biocontrol products, which can serve as alternatives to chemical pesticides to prevent disease spreading and crop loss. Although, substantial knowledge has been gained on genes influencing virulence, the distinction between genes specifically involved in entomopathogenic virulence versus general biological processes is insufficiently addressed. Previous studies have often focused on the role of global regulatory genes that control the expression of many downstream genes. Knocking-out a regulatory gene results in pleiotropic loss of fitness, and inevitably culminates in reduction of virulence on insects. Therefore the focus on dissecting the factors uniquely contributing to virulence on insects is a necessity. More specific virulence factors could be discovered via comparative gene expression analyses of *B. bassiana* strains with marked differences in virulence (Valero-Jiménez et al., 2014). Another strategy may be the use of an experimental evolution approach with the aim to identify genes and mechanisms underpinning virulence (unpublished data). Experimental evolution in combination with whole genome sequencing is a powerful tool since it could link genotype and phenotype (e.g. increased virulence, (Long et al., 2015)). An important additional aspect is that all conducted experiments should be based on a common methodology that comprises using similar target insects and a natural infection method with a constant concentration of spores and same time of exposure. Once specific virulence factors are identified, it will be important to explore the natural variation (in nucleotide sequence and expression of these genes) in the fungal population. It is also highly relevant to anticipate how insects can undergo adaptation to evade the action of fungal virulence factors and thereby develop resistance to the biocontrol agent.

Ultimately, these approaches could collectively increase the potential of biopesticides, through strain improvement in *B. bassiana*. It is also worth noting that other approaches that increase the potential applications of *B. bassiana* as a biocontrol agent have been suggested. For instance, transgenic strategies have been employed to express endogenous *B. bassiana* chitinases (Fang

et al., 2005), or heterologous toxins (Lu et al., 2008), or to express host molecules (Fan et al., 2012a), all in order to shift the balance of the interaction in favor of the pathogen, and thereby increasing efficacy in virulence. However, the use of genetically modified organisms has numerous hurdles that limit their applications in the field.

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